Fatim Majumder

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Applied Mathematics / Machine Learning PhD Applicant

Incoming Columbia M.S. in Applied Mathematics | Emory B.S. in Computer Science and Mathematics

Research areas: mathematical machine learning, scientific computing, optimization, statistical learning, AI for science, reliable AI systems

RESEARCH PROFILE

Applied mathematics and machine learning researcher focused on reliable empirical ML: reproducible evaluation, statistical model comparison, uncertainty-aware decision-making, and scientific computing infrastructure. My work connects proof-based mathematics with research engineering across LLM evaluation, biomedical ML, graph-based risk modeling, quantitative research, and robust state-space estimation.

Central question: How can modern machine learning systems be evaluated and improved with the rigor expected in numerical analysis, statistical inference, and scientific computation?

EDUCATION

Columbia University - M.S. in Applied Mathematics New York, NY | Incoming, expected start Fall 2026

Planned graduate focus: applied analysis, numerical methods, scientific computing, stochastic processes, optimization, statistical inference, computational modeling, and mathematical machine learning.

Emory University - B.S. in Computer Science and Mathematics Atlanta, GA | Expected May 2026

Selected coursework: Real Analysis I-II; Abstract Algebra I-II; Foundations of Mathematics; Linear Algebra; Multivariable Calculus; Differential Equations; Partial Differential Equations; Numerical Analysis; Numerical Optimization; Convex Optimization; Iterative Methods for Linear Systems; Mathematical Statistics I-II; Combinatorics; Machine Learning; Artificial Intelligence; Deep Learning on Graphs; Data Mining; Algorithms; Theory of Computing; Data Structures and Algorithms; Computer Architecture; Systems Programming.

Honors and activities: QuestBridge National Match Scholar; Dean's List; Undergraduate Research Distinction; HackDuke Finalist; Selective Technical Leadership Fellow; Varsity Rowing; Algory Capital Head of Research.

RESEARCH INTERESTS

- Mathematical ML: statistical learning theory, generalization, calibration, robustness, distribution shift, model selection, representation learning, graph learning, and evaluation methodology.
- Optimization and computation: stochastic approximation, adaptive first-order methods, mirror descent, momentum methods, convex and constrained optimization, conditioning, stability, and sensitivity analysis.
- Scientific computing and AI for science: numerical linear algebra, Krylov and iterative methods, simulation, state-space systems, biomedical ML, structure-aware learning, and uncertainty quantification.
- Reliable AI systems: LLM evaluation, benchmark design, regression detection, artifact lineage, inference orchestration, dataset versioning, observability, auditability, and failure analysis.

RESEARCH AND ENGINEERING EXPERIENCE

Arthur AI - Machine Learning Research Engineering Intern New York, NY | May 2025 - Aug. 2025

Technical stack: Python, PyTorch, Ray, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, AWS, OpenTelemetry, pytest, NumPy, pandas, scikit-learn, distributed queues, structured logging, object storage, benchmark orchestration.

- Built reproducible LLM evaluation infrastructure supporting 15,000+ benchmark runs per week across reasoning, retrieval, tool-use, multilingual, long-context, robustness, safety, and domain-specific suites.
- Formalized each evaluation run as a typed experimental object with explicit lineage over dataset snapshot, prompt template, tokenizer revision, model artifact, inference parameters, judge configuration, scoring function, postprocessor, aggregation rule, and execution environment.
- Increased exact rerun reproducibility from 74.1% to 99.96% with deterministic manifests, immutable artifact hashing, containerized evaluator images, schema-versioned prompts, seeded sampling controls, and provenance-preserving result storage.
- Designed asynchronous inference scheduling across heterogeneous model endpoints using adaptive batching, concurrency control, retry semantics, partial-failure isolation, and token-throughput accounting; reduced median experiment turnaround by 84% and wasted GPU-hours by 51%.
- Implemented slice-level regression detection using bootstrap confidence intervals, stratified score deltas, multiple-comparison-aware alerting, and run-diff reports, replacing aggregate-only dashboards with statistically interpretable model comparison tools.
- Found and corrected a preprocessing/de-duplication defect that inflated an internal benchmark by 6.1 absolute points; added invariant checks to prevent recurrence.
- Developed judge-calibration and uncertainty views for inter-judge agreement, rubric-level variance, drift checks, sampling sensitivity, and disagreement clustering.

Fullstory - Software Engineering Intern Atlanta, GA | May 2024 - Aug. 2024

Technical stack: Python, SQL, Kafka, Airflow, dbt, PostgreSQL, Docker, GitHub Actions, Grafana, OpenTelemetry, lineage metadata, anomaly detection, feature validation.

- Built validation infrastructure for analytics pipelines processing 4B+ weekly events, covering schema evolution, freshness lag, join integrity, replay mismatch, null spikes, metric discontinuities, cardinality explosions, and deploy-correlated anomalies.
- Reframed pipeline monitoring as statistical inference over changing data-generating processes, separating expected seasonality, product-driven distribution shift, instrumentation change, upstream lag, and genuine data-quality failures.
- Improved alert precision from 63% to 96%, reduced false positives by 79%, and lowered median time-to-detection from 3.4 hours to 7 minutes through layered validators and adaptive thresholds.
- Built anomaly attribution tooling linking metric shifts to deployment diffs, schema changes, ownership metadata, upstream delays, column-level distribution shifts, and lineage-neighbor failures.
- Implemented replay validation for event transformations, comparing historical and current outputs under controlled execution windows to detect non-idempotent transformations and hidden dependency changes.
- Reduced median debugging time from 2.9 hours to 24 minutes with a self-serve triage interface surfacing suspect fields, failed invariants, lineage paths, query diffs, and rollback context.

Algory Capital, Emory University - Head of Research Atlanta, GA | Sept. 2023 - Present

Technical stack: Python, pandas, NumPy, scikit-learn, XGBoost, statsmodels, SQL, PostgreSQL, DuckDB, Streamlit, FastAPI, Docker, GitHub Actions, Jupyter, backtesting engines, experiment tracking.

- Led research infrastructure and analyst development for a 30+ member quantitative research organization.
- Architected a point-in-time research platform spanning ingestion, universe construction, feature generation, cross-sectional factor analysis, walk-forward backtesting, portfolio optimization, exposure control, execution simulation, and attribution.
- Increased research throughput by 6.8x by replacing ad hoc notebooks with versioned experiment templates, canonical data loaders, reproducible tearsheets, automated validation checks, and standardized review rubrics.
- Implemented walk-forward model evaluation with time-aware splits, universe retention tracking, transaction-cost modeling, slippage assumptions, turnover constraints, benchmark comparison, and factor decay analysis.
- Built portfolio construction tools supporting mean-variance optimization, rank-weighted portfolios, sector/industry exposure constraints, volatility targeting, drawdown-aware diagnostics, and risk attribution.
- Added statistical validation layers for multiple testing, factor crowding, lookahead bias, survivorship bias, unstable correlations, data-snooping, and backtest overfitting.
- Mentored analysts on hypothesis formation, null models, ablation studies, leakage detection, uncertainty estimates, and separation of exploratory analysis from confirmatory testing.

Georgia Tech / Emory University - Research Assistant Atlanta, GA | Sept. 2022 - Jan. 2024

Wallace H. Coulter Department of Biomedical Engineering | Technical stack: Python, PyTorch, scikit-learn, NumPy, pandas, SQL, Streamlit, MATLAB, imaging preprocessing, clinical feature engineering, calibration analysis, statistical testing, experiment tracking.

- Developed biomedical ML pipelines supporting 1,000+ controlled experimental runs across preprocessing choices, cohort definitions, feature representations, multimodal fusion strategies, calibration procedures, threshold policies, and subgroup analyses.
- Reworked cohort construction to eliminate patient overlap, temporal contamination, feature leakage, and label-derived proxy variables, converting optimistic retrospective splits into clinically meaningful held-out validation.
- Improved held-out AUROC from 0.71 to 0.88 after correcting leakage, standardizing preprocessing, tuning model families, and introducing systematic ablations across imaging, clinical, and fused feature sets.
- Built subgroup evaluation reports measuring calibration, sensitivity, specificity, AUROC, AUPRC, threshold stability, and error modes across clinically meaningful strata.
- Implemented reproducibility infrastructure for dataset snapshots, preprocessing manifests, train/validation/test splits, model hyperparameters, random seeds, plots, metrics, and experiment notes.
- Designed ablation studies to distinguish genuine predictive signal from preprocessing artifacts, site-specific confounding, missingness patterns, and distributional shortcuts.

SELECTED COMPUTATIONAL RESEARCH PROJECTS

RE-AMP: Generative Audio Robustness Evaluation

Stack: Python, PyTorch, FastAPI, React, TypeScript, Redis, PostgreSQL, Docker, async workers, artifact lineage.

- Built a full-stack benchmarking platform for generative audio models under perturbations, distribution shifts, compression artifacts, prompt variation, acoustic transformations, and model/version changes.
- Supported 6,000+ controlled robustness runs with async job orchestration, structured artifact storage, deterministic run manifests, versioned benchmark suites, evaluator configuration, and comparative dashboards.
- Reduced experiment setup time by 89% with declarative benchmark specifications and reusable dataset/model adapters.
- Designed signal-level, perceptual, and model-based metric panels for robustness curves, failure slices, perturbation sensitivity, and ranking instability across model variants.
- Surfaced model-specific failure modes hidden by aggregate MOS-style summaries, including narrow-band degradation, prompt-template brittleness, and non-monotone compression response.

Graph-Based Traffic Collision Risk Modeling

Stack: Python, PyTorch Geometric, NetworkX, GeoPandas, SQL, XGBoost, scikit-learn, spatial statistics, graph learning.

- Built a graph ML pipeline over 1M+ collision records and road-network topology, representing intersections and road segments with spatial, temporal, traffic, and structural covariates.
- Formulated collision risk as heterogeneous spatial graph prediction and compared graph neural networks with strong tabular, geospatial, and time-windowed baselines.
- Improved hotspot-ranking AUROC by more than 20 points over tabular baselines by incorporating network connectivity, neighborhood structure, centrality features, temporal aggregation, and message passing.
- Implemented leakage-aware spatial and temporal validation splits to prevent nearby-edge contamination and future-information leakage.
- Added uncertainty and interpretability workflows: risk maps, feature ablations, graph-neighborhood explanations, calibration curves, error-slice analysis, and ranking stability under resampling.

Robust Kalman Filtering for Vehicle Localization

Stack: Python, NumPy, SciPy, MATLAB, C++, stochastic simulation, state-space models, numerical methods.

- Implemented a modular state-space simulation environment with configurable dynamics, observation models, process noise, measurement noise, dropout patterns, outlier regimes, delayed sensors, and asynchronous update schedules.
- Compared standard Kalman filtering with robustified variants using covariance inflation, residual gating, adaptive noise estimation, Huber-style updates, and outlier rejection.
- Reduced median localization error by 46% and 95th-percentile error by 41% relative to a standard Kalman filter under mixed dropout/outlier regimes.
- Analyzed filter breakdown under covariance misspecification, high-leverage measurement outliers, correlated noise, and delayed observations.
- Connected empirical results to recursive Bayesian estimation, linear-Gaussian assumptions, observability, stability, and numerical conditioning.

Quant Research Lab

Stack: Python, FastAPI, pandas, NumPy, SQL, Docker, GitHub Actions, Streamlit, statistical validation, backtesting.

- Built a public-facing factor research and backtesting demo covering point-in-time ingestion, signal construction, portfolio formation, evaluation, and attribution.
- Added leakage diagnostics for lookahead variables, survivorship effects, future price references, unstable universe definitions, duplicate securities, and time-window alignment errors.
- Implemented tearsheets showing factor returns, rank information coefficients, turnover, drawdown, exposure decomposition, transaction-cost sensitivity, and benchmark-relative performance.
- Designed the project as a teaching artifact for rigorous empirical research: every chart links to assumptions, data windows, transformations, and validation constraints.

TECHNICAL NOTES, RESEARCH ARCHIVE, AND PRESENTATIONS

HONORS, AWARDS, AND LEADERSHIP

- QuestBridge National Match Scholar - nationally selective scholarship recognizing academic achievement, leadership, and intellectual promise.
- Dean's List, Emory University - recognition for sustained high academic performance.
- Undergraduate Research Distinction - recognition for sustained research engagement and scholarly contribution.
- HackDuke Finalist - finalist recognition for technical prototyping, applied problem solving, and interdisciplinary project execution.
- Selective Technical Leadership Fellow - selected for technical leadership, mentorship, software execution, and project development.
- Head of Research, Algory Capital - led research infrastructure and analyst development for a 30+ member quantitative research organization.
- Varsity Rowing - collegiate athletics; disciplined training, teamwork, endurance, and high-performance execution.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, TypeScript, JavaScript, Java, Bash, MATLAB.

Machine Learning and AI: PyTorch, PyTorch Geometric, scikit-learn, XGBoost, Ray, LLM evaluation, graph ML, multimodal ML, transformer inference, retrieval evaluation, calibration, robustness testing, ablation studies, uncertainty estimation, generative model evaluation.

Applied Mathematics and Scientific Computing: Numerical linear algebra, iterative methods, convex optimization, numerical optimization, stochastic approximation, state-space models, Kalman filtering, Monte Carlo simulation, statistical inference, spectral methods, graph algorithms, computational modeling.

Data and Research Infrastructure: PostgreSQL, MySQL, DuckDB, Redis, Kafka, Airflow, dbt, ETL, feature validation, dataset versioning, model/artifact lineage, experiment tracking, reproducible replay, structured logging.

Backend and Systems: FastAPI, Docker, Kubernetes, AWS, Linux, Git, GitHub Actions, CI/CD, distributed queues, asynchronous orchestration, caching, batching, observability, profiling, reliability engineering.

Evaluation and Reliability: Benchmark design, metric validation, regression detection, prompt and dataset lineage, statistical comparison, confidence intervals, run-diff tooling, audit logging, failure analysis, incident triage, data-quality testing.